9.1 Caring For Your Disk: Defragmentation

You can't do anything about a hard disk crash—or can you? Yes and No. The fact is, hard disks have mechanical parts, and will ultimately crash—and you can't predict when one will. You'll sometimes find a decade-old hard disk running perfectly fine, and you'll sometimes see a year-old hard disk that crashed for no reason at all. On the other hand, there are a few things you can do to prolong the life of your hard disk:

- Keep it running cool (which we'll come to later in this chapter)
- Prevent thrashing: don't make the hard disk access too many files at the same time. For example, don't play a song while a movie is already running.
- Set the disk to turn itself off within a few minutes of inactivity. The downside is that you'll have to wait a few seconds when you access a file after that time, but it's worth it. It prevents overheating and reduces wear and tear.
- Don't do anything that would shake up the mechanical parts—for example, frequently transporting it between two computers. Once in a while (if done with care) is OK, but don't make it a habit.
- Defragment it often. This, too, reduces wear and tear.

Why defragmenting needs to be done is essentially so that the head doesn't have to move "back and forth" too much.

Over a period of time, your files tend to get fragmented—one part of a file will be stored on one location on the disk, and the other part somewhere else. And it's not limited to two locations: there could be hundreds of fragments of one file all over the disk! (For a visual representation of this, see *How Disk Defragmentation Works*, *Digit,* December 2005.) Accessing all the fragments in sequence means the head will need to shuttle between the locations at high speeds. The more of this it needs to do, the more susceptible it is to losing its alignment—which, of course, means a crash.